



Onboarding Databases to DSF Hub Reference Guide

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This topic reviews the steps required to audit activity on Amazon DynamoDB databases and send that information to DSF Hub for auditing, analysis, detecting and preventing security events.

Getting Started

This page contains the information necessary to onboard this data source to Data Security Fabric (DSF) Hub. The following main topics are included:

- An overview of the available scope for the data source, given its native audit data.
- A complete list of prerequisites and permissions that are required for onboarding data sources to DSF Hub.
- Instructions on how to enable audit on the data source and collect data using DSF Hub.
- Initial troubleshooting steps and technical support information.

Onboarding Steps

To ensure a smooth and successful deployment, it is necessary to complete each of these onboarding steps. Please click on each of the steps below to display the content.

✓STEP 1: Onboarding Prerequisites

Onboarding a data source requires preparation, such as gathering permissions and collecting relevant information for your deployment. Assistance may be necessary from a database administrator, network administrator, and an IT administrator to successfully begin monitoring your data source.

Please ensure all of the following items are properly configured or available for use. The information and permissions gathered in this step are required for the remaining onboarding steps.

Cloud Account Prerequisites

Below are the requirements for an Agentless Gateway to communicate with data sources and their related endpoints. Please ensure these are completed by a Cloud Administrator.

Grant the AWS principal the following permissions:

- **AWS`CloudTrail_FullAccess` policy:** for creating and managing the CloudTrail, S3 bucket and log group in CloudWatch. Please refer to the [AWS CloudTrail](#) documentation for more details.
- **Amazon`EventBridgeFullAccess` policy:** for creating an EventBridge rule to forward the logs to CloudWatch.

AWS Cloud Account authentication mechanisms

DSF supports the following authentication mechanisms for accessing AWS resources:

- Key pair: for more information on creating a key pair, see [Manage access keys for IAM users](#).
- IAM roles: for more information on creating an IAM role, see [IAM roles for Amazon EC2](#).
- Profile
- Default

The authentication mechanism assigned to the Agentless Gateway should be granted the following privileges:

```
logs:DescribeLogGroups
logs:DescribeLogStreams
logs:FilterLogEvents
logs:GetLogEvents
```

✓STEP 2: Enabling Audit on the Data Source

Once the required permissions and information have been obtained, please complete the following steps to enable audit on the data source.

Enable Audit

Amazon DynamoDB supports auditing for both Management events and Data events:

- Management events, i.e. control plane operations, provide visibility into management operations that are performed on resources in your AWS account. By default, trails and event data stores are configured to log Management events.
- Data events, i.e. data plane operations, provide visibility into the resource operations performed on or within a resource. Data events are not logged by default. To capture Data event logging, a CloudTrail that exports the logs to an S3 bucket is required.

Note:

"Management events" are logged on the account level, so if at least one CloudTrail within the same region has "Management events" enabled, your CloudWatch log group will also receive the logs. If you don't need the "Management events", make sure that no other CloudTrail has this option enabled.

To ensure both Management and Data events are logged, the Eventbridge is used to forward management and data plane events to CloudWatch, from which the Agentless Gateway can retrieve the logs. For more details, see <https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/logging-using-CloudTrail.html>

Please follow the instructions below to create the CloudTrail for logging DynamoDB activity, either via the AWS Portal or AWS CLI.

Log DynamoDB Activity with EventBridge (via AWS Portal)

1. Creating a CloudTrail:

Create a new CloudTrail to enable logging for DynamoDB. This can be setup to audit only specific tables or all tables current and future. For more details, see [AWS documentation](#).

- Navigate to CloudTrail console and click on the "Create Trail" button.
- Enter "Trail name".
- To store the trail's event logs, in the Storage location, select Create new S3 bucket (or Use existing S3 bucket).
- For Trail log bucket and folder, enter an S3 bucket name.
- Click on "Next" at the end of the page.

2. Enabling logging:

- On the Choose log events page, choose Data events and Management events.
- In "Data Events type" click on "Switch to basic event selectors" button.
- For Data event source, choose DynamoDB from the dropdown
- (Optional) if you deselect Log data for all current and future DynamoDB tables you can configure the CloudTrail to only impact individual tables.
- Click on "Next"
- On the Review and create page, review the configuration and choose Create trail.

The next step will describe exporting all events (Data and Management) from an S3 bucket to CloudWatch.

3. Forward CloudTrail activity to LogGroups with EventBridge:

- To forward CloudTrail activity to log groups, create an EventBridge rule that matches an event pattern for events coming from DynamoDB.

4. Create an EventBridge Rule:

- From the EventBridge console, click "Create rule" on the far right.
- Specify a "Name"
- For "Rule Type" choose "Rule with an event pattern"
- Click on "Next"
- For "Event Source", choose "Other"
- In "Creation method", choose "Custom patterns (JSON editor)" and in "Event pattern" add the following code and click "Next".

```
{
  "source": ["aws.dynamodb"],
  "detail": {
    "eventSource": ["dynamodb.amazonaws.com"],
    "sourceIPAddress": [{
      "anything-but": ["replication.dynamodb.amazonaws.com", "dynamodb.application-autoscaling.amazonaws.com"]
    }]
  }
}
```

- For "Target types" select "AWS Service" and choose "CloudWatch log group" from the dropdown.
- Provide a name for the log group.
- Click on Next → Next.
- Review the details and click on "Create rule".

For more information on AWS EventBridge filtering, see: <https://docs.aws.amazon.com/eventbridge/latest/userguide/what-is-amazon-eventbridge.html>.

Generate traffic on your AWS DynamoDB source and check that the logs are being sent to your CloudWatch log group.

Log DynamoDB Activity with EventBridge (via AWS CLI)

To log DynamoDB activity, create an EventBridge rule matching specified incoming events - DynamoDB API calls, and route them to an EventBridge target (CloudWatch Log Group) for processing. For more information on the AWS resources that can act as targets for EventBridge, see [Amazon EventBridge documentation](#).

1. Create a Trust Policy:

To allow EventBridge to write the audit events to the CloudWatch log group, create a service role with a policy and attach it. On the machine where AWS CLI is installed, create a local file called EventBridge-Role-Trust-Policy.json with the following code to create the trust policy allowing the EventBridge service to assume roles.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": {
        "Service": "events.amazonaws.com"
      },
      "Action": "sts:AssumeRole"
    }
  ]
}
```

2. Create a Service Role:

Create a service role for EventBridge by attaching the trust policy from the previous step.

```
aws iam create-role \
--role-name EventBridge-Role \
--assume-role-policy-document file://EventBridge-Role-Trust-Policy.json
```

Make note of the role arn:

```
"arn:aws:iam::{{account}}:role/EventBridge-Role"
```

3. Create a Cloudwatch Policy:

In a local file called EventBridge-To-CloudWatch-Policy.json, define the permissions allowing EventBridge to write to CloudWatch log groups. Be sure to replace **region** and **account** with the correct values:

```
{
  "Statement": [
    {
      "Action": [
        "logs:CreateLogStream",
        "logs:PutLogEvents"
      ],
      "Effect": "Allow",
      "Resource": "arn:aws:logs:{{region}}:{{account}}:log-group:/aws/events/*:*",
    }
  ]
}
```

```

        "Sid": "TrustEventsToStoreLogEvent"
    },
    ],
    "Version": "2012-10-17"
}

```

Create the policy with the permissions file above:

```

aws iam create-policy \
--policy-name EventBridge-To-CloudWatch-Policy \
--policy-document file://EventBridge-To-CloudWatch-Policy.json

```

Make note of the policy's arn:

```

"arn:aws:iam::{{account}}:policy/EventBridge-To-CloudWatch-Policy"

```

4. Create an EventBridge Role:

Attach the policy arn to the service role's friendly name (not role arn):

```

aws iam attach-role-policy \
--role-name EventBridge-Role \
--policy-arn "arn:aws:iam::{{account}}:policy/EventBridge-To-CloudWatch-Policy"

```

5. Create an EventBridge Rule:

Create an EventBridge rule that filters for DynamoDB events, specifying the role (arn) to use:

```

aws events put-rule \
--name sonar-dynamodb-rule \
--event-pattern \
'{
  "source": ["aws.dynamodb"],
  "detail": {
    "eventSource": ["dynamodb.amazonaws.com"],
    "sourceIPAddress": [{
      "anything-but": ["replication.dynamodb.amazonaws.com", "dynamodb.application-autoscaling.amazonaws.com"]
    }]
  }
}' \
--state ENABLED \
--role-arn "arn:aws:iam::{{account}}:role/EventBridge-Role"

```

For more information on AWS EventBridge filtering, see: <https://docs.aws.amazon.com/eventbridge/latest/userguide/what-is-amazon-eventbridge.html>

6. Add a Target to the EventBridge Rule:

The audit events should be sent to a Cloudwatch Log group via the Eventbridge rule. Use an existing Log group or create a new one by running the following command:

```

aws logs create-log-group \
--log-group-name /aws/events/sonar-dynamodb-log-group

```

Add the CloudWatch log group as a target to the EventBridge rule by running the following command:

Note:

The log group should already exist/be created before adding it to the rule.

```
aws events put-targets \
--rule sonar-dynamodb-rule \
--targets "Id"="sonar-target-id", "Arn"="arn:aws:logs:{{region}}:{{account}}:log-group:/aws/e
```

7. Generate traffic on your AWS DynamoDB source and check that the logs are being sent to your CloudWatch log group.

✓STEP 3: Collecting Audit Data

After completing the prerequisites and enabling audit, the data source is ready to be onboarded onto DSF. This can be accomplished using **ANY ONE** of the methods listed below:

- Importing Assets via Unified Settings Console (USC)
- Importing Assets via an Asset Template Spreadsheet
- Importing Assets via DSF Open APIs

Please use the Asset Specifications documents below as a guide to fill in the field values for this data source and if applicable, its related assets.

[AWS Asset Specifications](#)

[AWS DynamoDB Asset Specifications](#)

[AWS Log Group Asset Specifications](#)

Note:

When onboarding an Amazon data source that exports audit data via **Amazon CloudWatch**, three assets are required:

- An **AWS Cloud Account** asset
- A **Database or Instance** asset
- An **AWS Log Group** asset

Define the relationship between these assets using the `parent_asset_id` field (or **Parent Asset Name** when using USC). The asset hierarchy should be as follows:

- The **AWS Cloud Account** is the parent of the **Database or Instance** asset.
- The **Database or Instance** asset is the parent of the **AWS Log Group** asset.

Refer to the asset template spreadsheet for a visual demonstration of the asset structure and relationship.

For download instructions, see [Downloading Asset Templates](#).

Importing Assets via Unified Settings Console (USC)

USC allows users to configure a full audit flow, including importing new data assets. To add a new data source asset in USC, please follow these steps:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.
2. Click the Data Sources tab to view the data sources of the DSF Hub. Click **Add** to open the Add Data Source form.
3. In the Data Source Type dropdown menu, select a data source.
4. Specific data source configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.
5. If required for the data source, under Connection, select an authentication method (Auth Mechanism) from the dropdown menu. The mandatory fields for the selected Auth Mechanism are displayed; to see optional configuration fields, click Advanced.
6. Under Monitoring, select the Agentless Gateway that will own the data source in the Gateway field, then click **Save**.
7. Locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on adding, viewing and editing assets in USC, see the [Adding Assets via Unified Settings Console \(USC\)](#) documentation.

Importing Assets via an Asset Template Spreadsheet

Complete these steps to import data source assets using asset template spreadsheets:

1. From the DSF Portal of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.
3. Use the Asset Specification documentation as a guide for filling in the necessary field values.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **'Validate All'**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the data source asset that was imported.

8. Click the kebab menu to the right of the asset, then click **Management > Audit Actions > Connect Gateway** to start collecting audit data.

For more information, please see the [Adding Assets via the Import Assets Page](#) documentation.

Note:

Use the following asset templates when importing via spreadsheet:

- Asset template spreadsheet name for Amazon DynamoDB: ***AMAZON_DYNAMODB_template.xlsx***
- Asset template spreadsheet name for AWS account; choose one of the four templates listed below depending on the type of authentication mechanism:
 - ***AMAZON_EC2_DEFAULT_ROLE_template.xlsx***
 - ***AMAZON_IAM_ROLE_template.xlsx***
 - ***AMAZON_KEY_PAIR_template.xlsx***
 - ***AMAZON_PROFILE_template.xlsx***

Importing Assets via DSF Open APIs

Data Security Fabric (DSF) Open APIs provide functions for onboarding and managing assets (log aggregators, cloud accounts, data sources, secret managers and other assets) via a RESTful API. For more information on the supported assets and how to onboard them, please see [Using DSF Open APIs](#).

Troubleshooting

Should you encounter any unexpected issues or behaviors, you may check the status of the following services and associated log files to help pinpoint the root cause. If additional assistance is needed at any time, technical support staff is available to help users of all technical levels via <https://supportportal.thalesgroup.com/csm>.

On the Agentless Gateway host(s) please review the following:

- Gateway Log file: `$JSONAR_LOGDIR/gateway/cloud/aws/dynamodb/sonargateway.log`
- Run the following command to verify the status of the gateway service:

```
systemctl status -l gateway-aws@dynamodb.service
```

For more information...

Need Help? For assistance with any DSF Hub or related products, please contact Online Technical Support via <https://supportportal.thalesgroup.com/csm>. A team of technical customer success representatives are ready to assist users of all skill levels.

Related Topics: Below are links with related onboarding information and procedures:

- [Using DSF Open APIs: A guide to onboarding and managing assets via DSF Open APIs.](#)

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- [Adding assets via Unified Settings Console \(USC\)](#): Instructions to add and edit data source assets via the USC.
 - [Adding Assets via the Import Assets Page](#): Instructions to add data source assets via the Asset Dashboard.
 - [Using the Data Security Fabric Portal](#): An overview of the Data Security Fabric (DSF) portal.
 - [Data Security Coverage Tool](#): Systems and version compatibility with DSF Hub.

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